Introduction to Data Science

Outline



What is Data Science?



Where does data come from?

Why the excitement?



Machine Learning Models

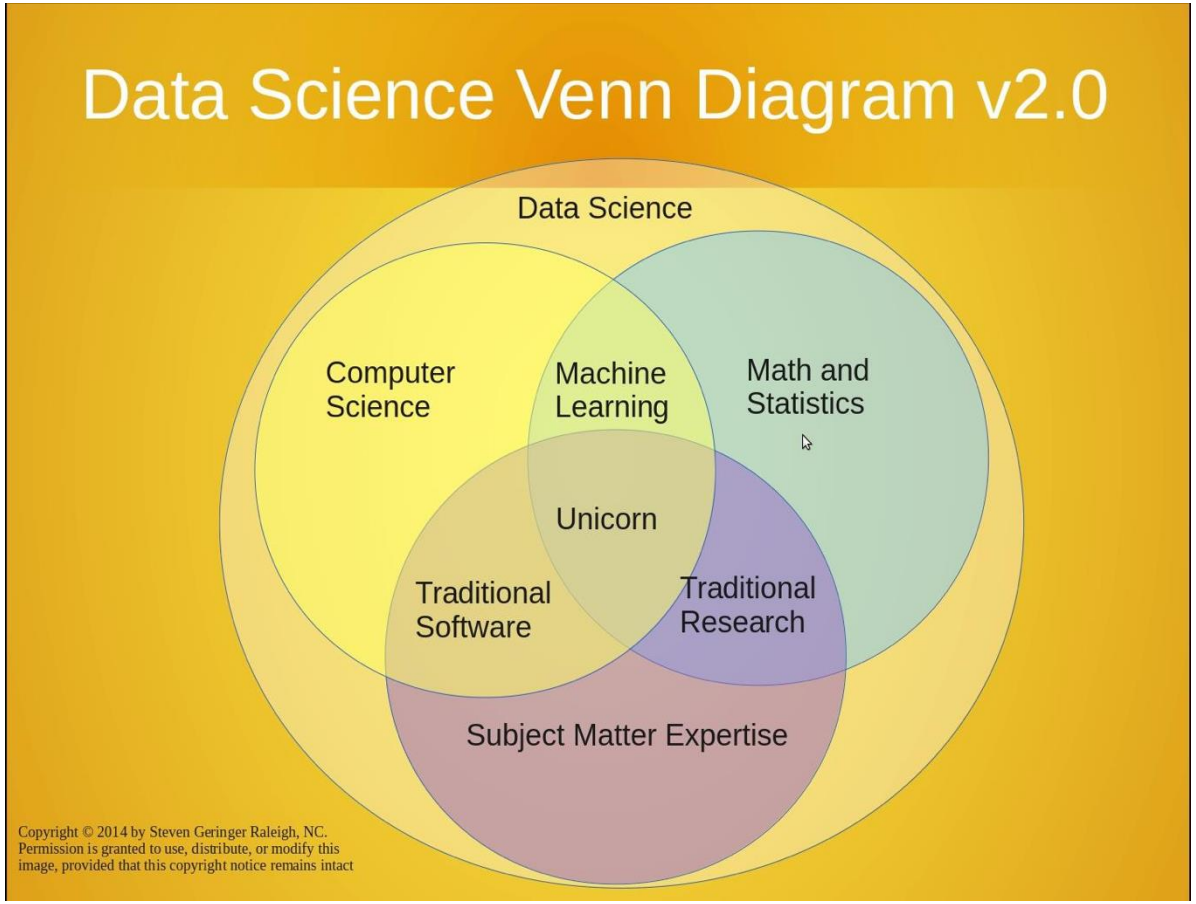


The Data Science Process

What is Data Science?

History & definition

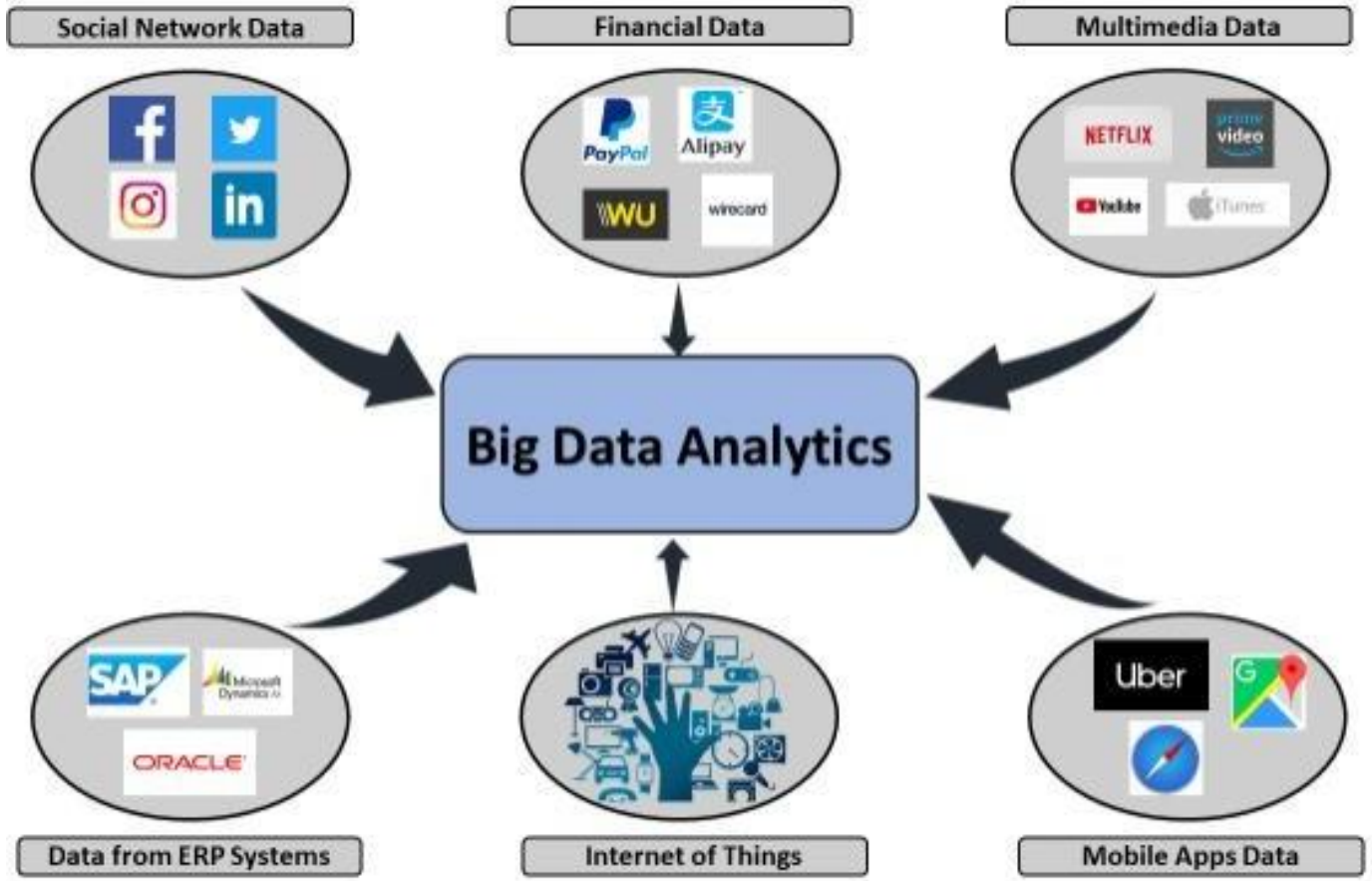
Data Science Definition



Data Science is an art of analyzing and extracting knowledgeable information from the data.

Where does Data come from?

Big Sources of Data



Why the Excitement?

Data Science in Education



https://www.pomona.edu/news/2026/03/11-new-data-science-minor-helps-students-learn-skills-solve-wide-ranging-problems

Data Science in Malaysia

SAS collaborates with the Government to advance Malaysia's Data Analytics and AI Capabilities

The announcement was made at SAS' new office opening graced by YB Gobind Singh Deo

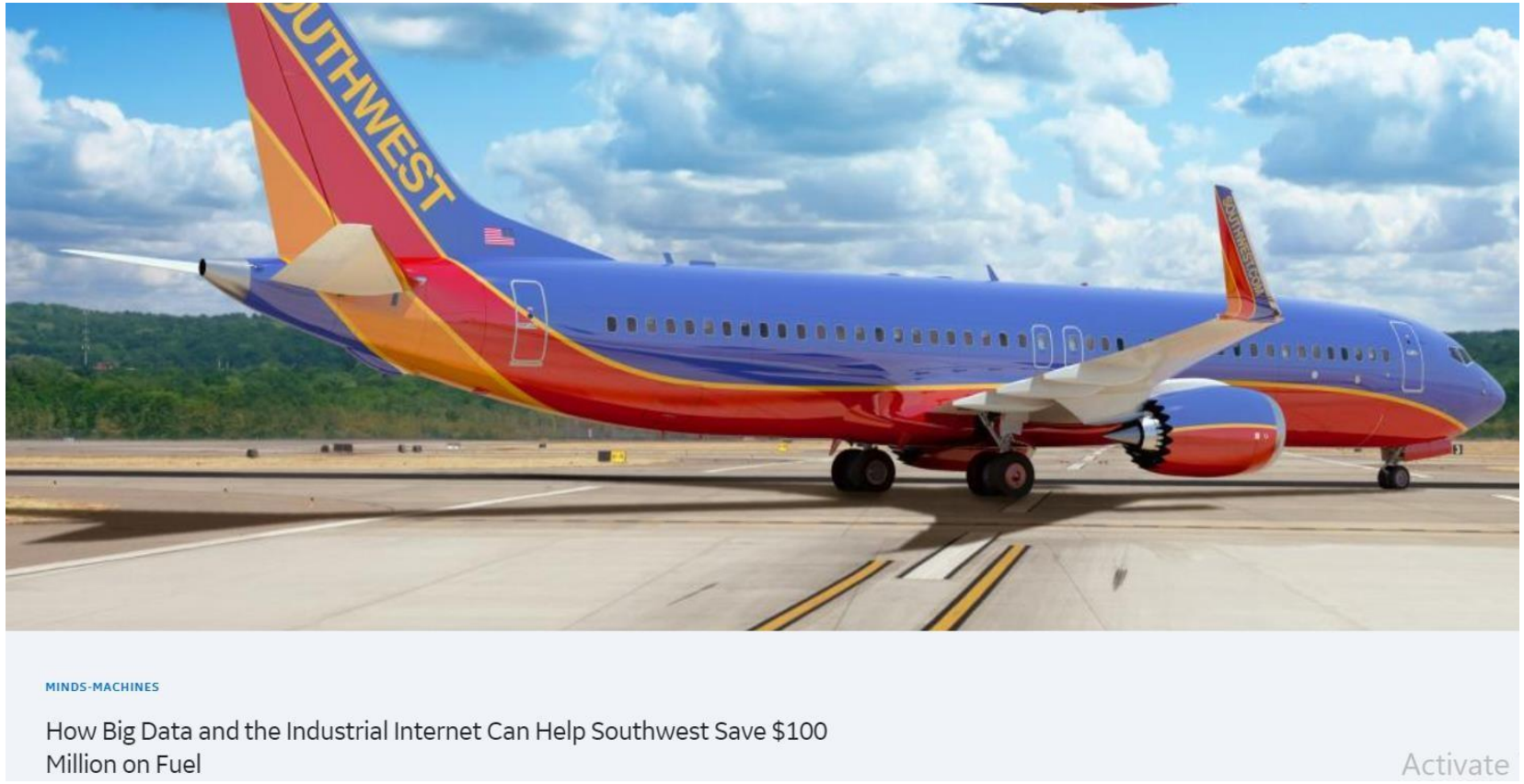


Editorial contacts:

Cherie Ho
+65 6398 3385

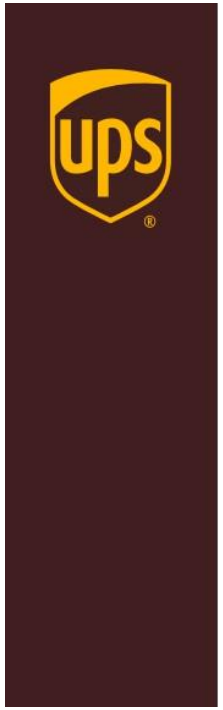
https://www.sas.com/en_my/news/press-releases/2024/september/sas-collaborates-with-the-government-to-advance-malaysia-data-analytics-and-ai-capabilities.html

How Data Science Helped Southwest Save $100 Million on Fuel?



[How Big Data and the Industrial Internet Can Help Southwest Save $100 Million on Fuel | GE News]

How Data Science helped UPS save 39 millions of Gallon by route optimization?

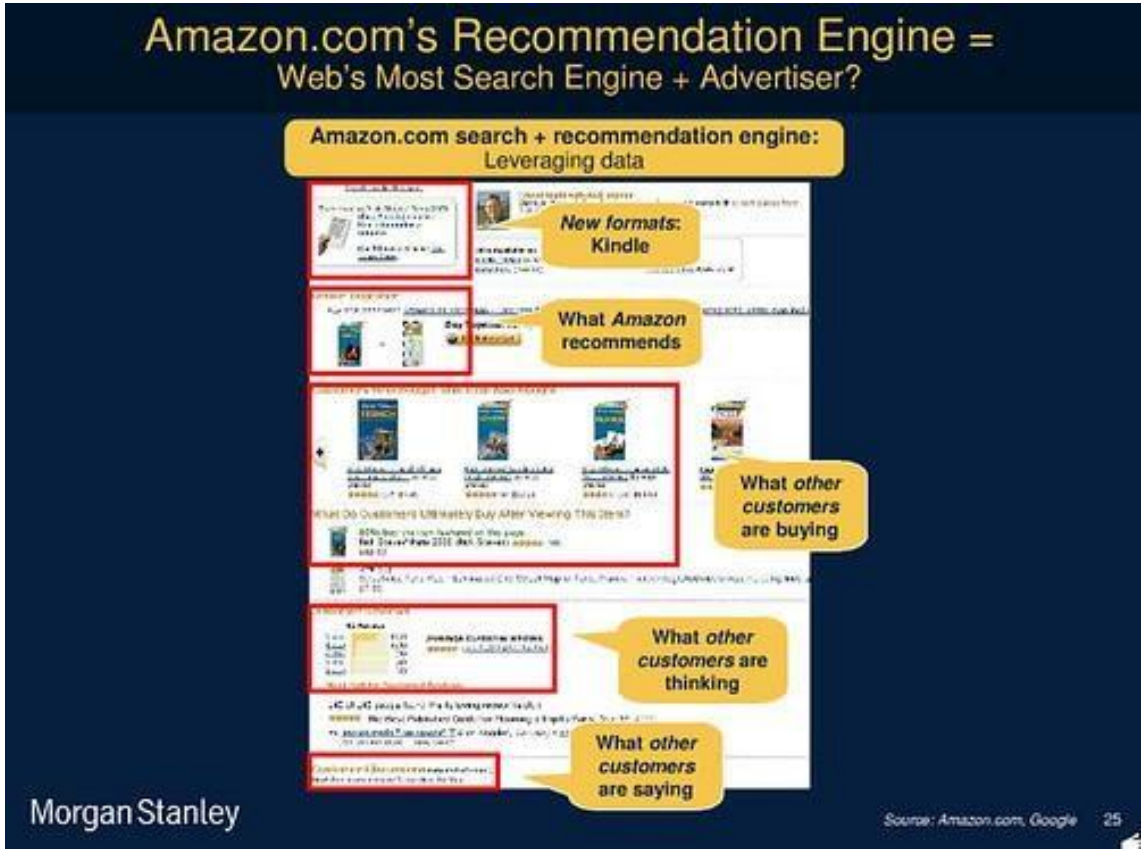


Big Data = Big Wins for the Environment

UPS manages more than 16 million shipments a day, all over the world. Its single integrated and optimized network results in environmental benefits such as reductions in fuel use and greenhouse gas emissions. To make the network operate more efficiently and reduce environmental impact, UPS designs, acquires, implements and optimizes information technology for continuous improvement. Technology has helped UPS fine-tune various aspects of its operations – from planning and routing to flying and driving – something that's good for business and the environment.

big-data-infographic_v06 (ups.com)

Amazon Recommendation System



Amazon - recommendation engine | Youssef Rahoui | Flickr

Netflix Movie Suggestion





Netflix's Billion-Dollar Secret: How Recommendation Systems Fuel Revenue and Innovation

https://www.linkedin.com/pulse/netflixs-billion-dollar-secret-how-recommendation-systems-qin-phd-7zece/

US Polling: 2008 & 2012

Nate Silver uses a simple idea – taking a principled approach to aggregating polling instead of relying on punditry – and:

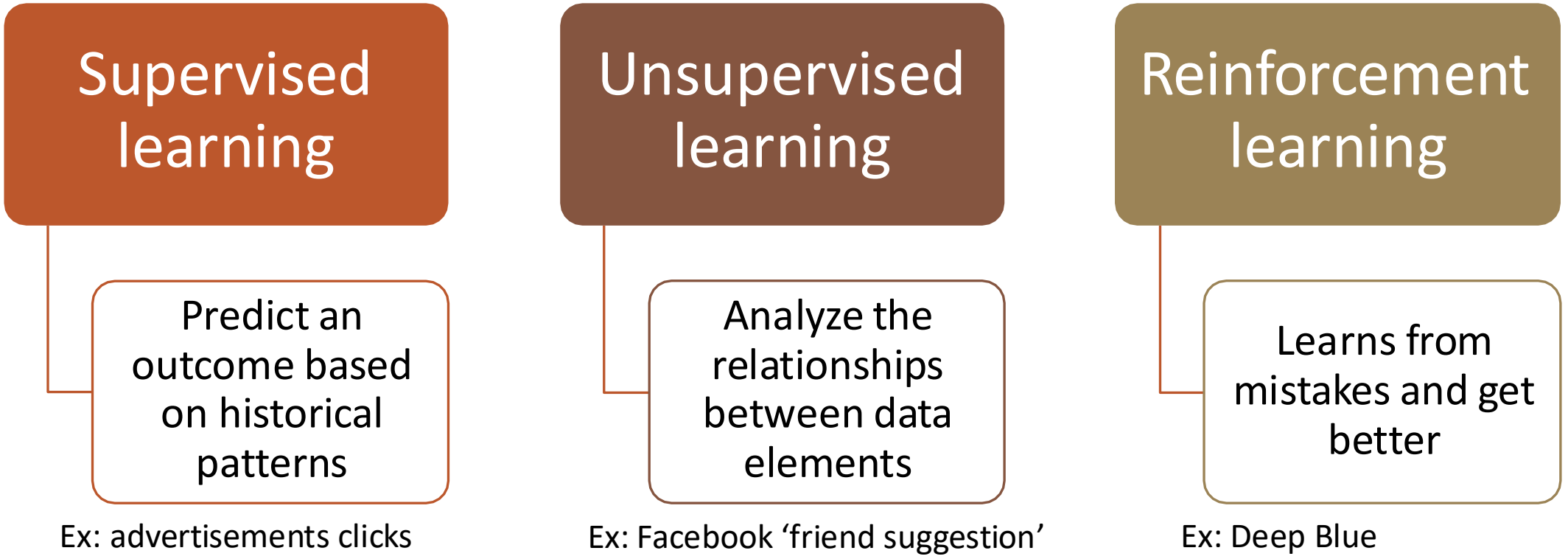
- Predicts 49/50 states in 2008
- Predicts 50/50 states in 2012



https://hbr.org/2012/11/how-nate-silver-won-the-2012-p

Machine Learning Models

Machine Learning models



The Data Science Process

The Data Science Process


Formulate the question


Generate Hypothesis


Collect the data


Clean Data


Explore/Transform Data


Build Machine Learning models


Evaluate & Deploy the Model

Formulate the question

- Understand the business problem

- Excellent results depend on a better understanding of the problem.

Generate Hypothesis

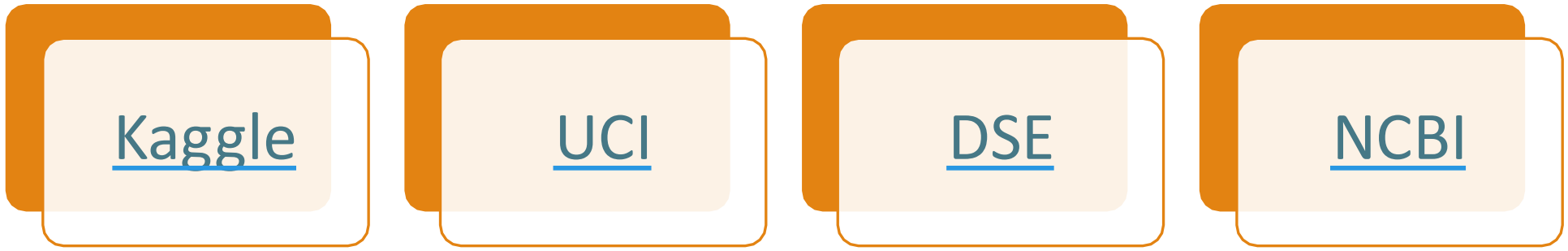
- Guess an approach through which we derive some essential data parameters that have a significant correlation with the prediction target.

- Let’s take an example of loan approval prediction, some features:

  - **Income:** If an applicant has a higher income, should get a loan easily.
  - **Education:** Higher education results in higher income, so we can approve the loan request.
  - **Loan Amount:** Lesser the amount, chances of loan approval are high.

Collect the data

Gathering data from relevant sources!



Kaggle: Your Machine Learning and Data Science Community

UCI Machine Learning Repository

Dataset Search (google.com)

National Center for Biotechnology Information (nih.gov)

Clean Data (Data Wrangling)



5 Data Science Projects That Will Get You Hired in 2018 - KDnuggets

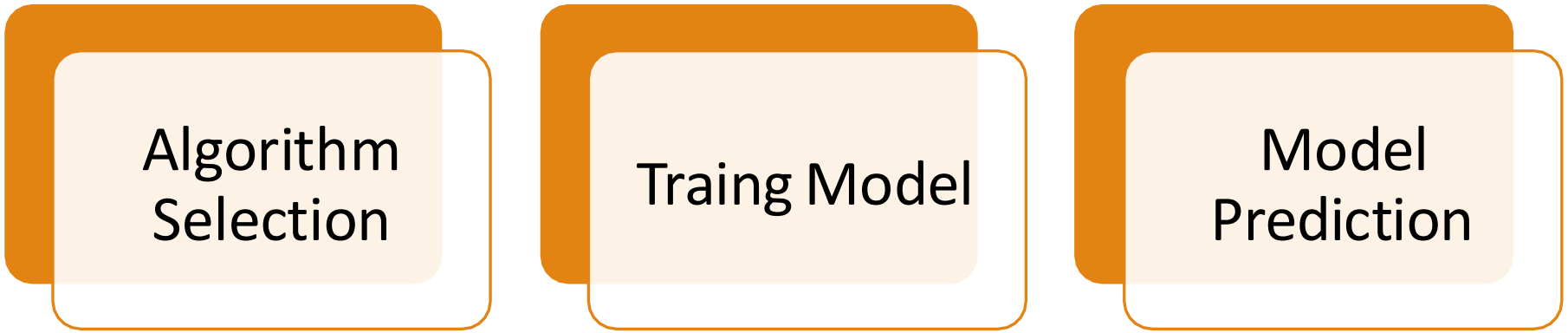
Some common source of data errors:

- Duplicate entries
- Missing values

Explore/ Transform Data

- Feature Identification
- Univariate Analysis
- Multi-variate Analysis
- Feature Engineering

Build Machine Learning Models



Evaluate & Deploy the Model

Validate selected Machine learning model

Deploy model and make it accessible through an API

Can you plan your own data science project?

Q & A?